

Conditional Survival Analysis

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We aim to get an understanding of how conditional survival works on judicial data.
We will look at:

- ▶ Conditional survival and its relevance;
- ▶ How survival probabilities are derived given a condition;
- ▶ Example of conditional survival on hearings;
- ▶ Further research.

Key Takeaways

- ▶ Conditional survival refines predictability by giving a more precise probability.
- ▶ If a case has an early first hearing, it is more likely to progress faster through the system.
- ▶ Dropping cases that haven't had an event yet skews survival probabilities.

Survival Analysis

Survival analysis is used to analyze the time until an event of interest occurs.

Concepts:

- ▶ Event: The outcome of interest;
- ▶ Time: The duration from a starting point to ending point;

Survival Analysis

Survival analysis is used to analyze the time until an event of interest occurs.

Concepts required:

- ▶ Event: The outcome of interest
- ▶ Time: The duration from a starting point to ending point
- ▶ **Censoring:** Incomplete observation of the event. Instead of truncating the data, it incorporates missing information or if the subject did not occur within a specific time period.

Conditional Probability

Conditional probability is the probability of an event occurring given that another event has already occurred.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (1)$$

- ▶ $P(A|B)$: Probability of event A occurring given event B
- ▶ $P(A \cap B)$: Probability of both A and B occurring
- ▶ $P(B)$: Probability of event B occurring



Conditional Survival

Having said the above, **conditional survival** is the probability of surviving an additional period of time, given that an individual has already survived up to a certain point.

- ▶ In terms of judicial data, we would be willing to find the probability of a court case attaining an event (say getting further hearing), conditioned on another event (say getting an initial hearing) occurring
- ▶ Conditional survival provides a more targeted probability estimate and analyse temporal dependencies between events. This translates to a concept called Kaplan Meier estimates

Conditional Kaplan Meier Estimate

- ▶ This is better addressed as a sub-part of conditional survival
- ▶ Conditional Kaplan Meier estimate provides a targeted probability, accounts for censoring (refer to slide 4) and gives us unbiased results
 - ▶ The timing of the first hearing affects the timing of the second hearing. By conditioning on the first hearing occurring within the first year, you account for this temporal dependence and provide a more accurate estimate of the probability of the second hearing
- ▶ Bias arises when we truncate/drop data

What is the bias?

Truncation leads to a selection bias:

- ▶ If we drop off cases that haven't yet had the event (e.g., first hearing being conditioned), we ignore valid observations that are still in progress.
- ▶ In addition, if we only analyze cases that have reached the event, our estimates of conditional survival probabilities will be too optimistic.

Survival Function:

$$S(t) = P(T > t) \quad (2)$$

$S(t)$ gives us the probability that a case survives (given any event) beyond a certain time point t

Conditional Survival Function:

$$CS = S(t|T_0) = \frac{S(T_0 + t)}{S(T_0)} \quad (3)$$

- ▶ CS : Conditional Survival
- ▶ $S(t|T_0)$: Probability of surviving additional t years given survival up to time T_0 .
- ▶ $S(T_0 + t)$: Overall survival probability at time $T_0 + t$.
- ▶ $S(T_0)$: Overall survival probability at time T_0

Questions Answered through Sample Question

- ▶ Does an early first hearing lead to a faster second hearing?
 - ▶ If cases with early first hearings also get second hearings quickly, it suggests **efficient case processing**.
- ▶ What is the typical wait time between the first and second hearing for cases that had their first hearing within 3 months?
 - ▶ It can help courts to set benchmarks to address delays

Sample Question

- ▶ Scenario: A case getting its second hearing within a specific point of time t from the date of the first hearing, given it had got its first hearing within 3 months.
- ▶ Dataset: Cases filed between January 2021 and December 2023.
- ▶ Variables of interest: time to first hearing, time to second hearing, and the event (whether the case had a second hearing and the first hearing happened within 3 months).

Sample Question

Statistically,

$$P(T_2 < t | T_1 < 0.25) = \frac{P(T_1 < 0.25 \cap T_2 < t)}{P(T_1 < 0.25)} \quad (4)$$

- ▶ T_1 : Time to first hearing
- ▶ T_2 : Time to second hearing
- ▶ t : Any given time point of consideration
- ▶ $P(T_2 < t | T_1 < 0.25)$: Probability of getting a second hearing within t given the first hearing occurred within 0.25 years (3 months)



Sample Question Results

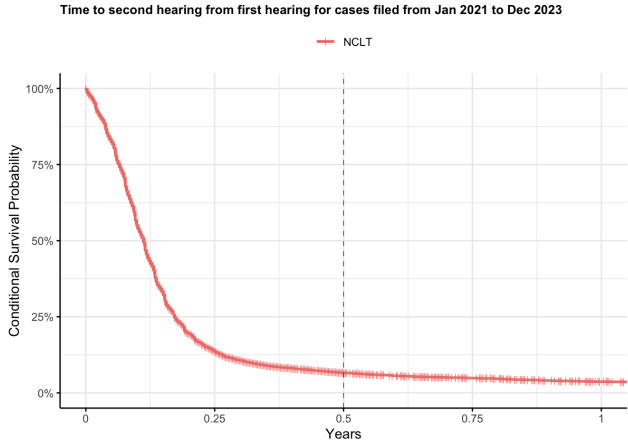


Figure: Time to second hearing from first hearing with condition

Sample Question Results

Values for any given time point t :

When $t = 0.25$ (3 months), 0.5 (6 months), 0.75 (9 months), 1 (1 year):

"Cases getting second hearing within next 3 months , given the first hearing happened within 3 months : 86.3 %"

"Cases getting second hearing within next 6 months , given the first hearing happened within 3 months : 93.3 %"

"Cases getting second hearing within next 9 months , given the first hearing happened within 3 months : 95.1 %"

"Cases getting second hearing within next 1 year , given the first hearing happened within 3 months : 96.3 %"

Time to Second Hearing - Without Conditioning

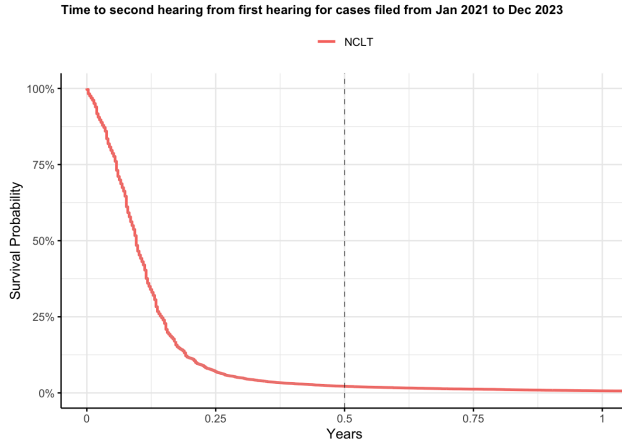


Figure: Time to second hearing from first hearing without being conditioned

Results without Conditioning

Values for any given time point t :

When $t = 0.25$ (3 months), 0.5 (6 months), 0.75 (9 months), 1 (1 year):

"Cases getting its second hearing within 3 months of first hearing: 93 %"

"Cases getting its second hearing within 6 months of first hearing: 97.8 %"

"Cases getting its second hearing within 9 months of first hearing: 98.8 %"

"Cases getting its second hearing within 1 year of first hearing: 99.3 %"

Comparison of Time to First and Second Hearing from Filing

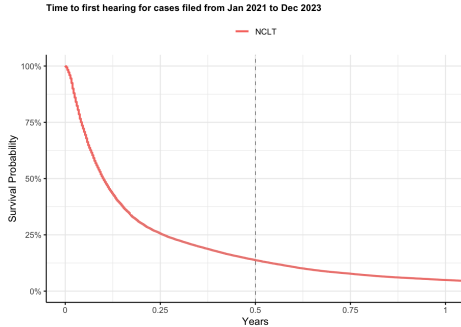


Figure: Time to first hearing from filing a case

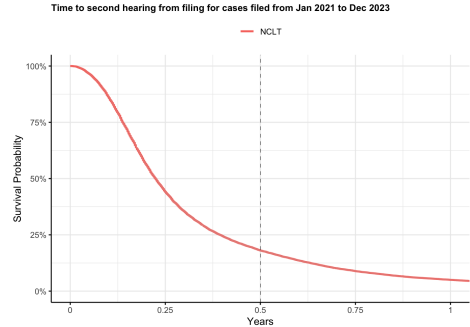


Figure: Time to second hearing from filing a case

Further Research

- ▶ While conditional survival (using Kaplan Meier) is non-parametric, further research can be done to look into the possibilities of including parameters/factors affecting.
- ▶ This comes from using a cox proportional model. This gives a true and better picture of the underlying data.

References

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3. Stefanie H; Martina K; Christine K; Monika E; Martin S. Conditional Survival: A Useful Concept to Provide Information on How Prognosis Evolves over Time. AACR. 2015

Thank you
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